

TICVAI

AI Fraud & Risk Intelligence

Dashboard requirements and functional specification

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AI Fraud & Risk Intelligence — Board 1: Payment, Transaction, Refund & Chargeback Fraud

Module: AI Fraud & Risk Intelligence

Board: 1 of 3

Screens: 10

Board Type: Fraud Detection / Transaction Risk / Payment Intelligence / Refund Abuse / Chargeback Intelligence



Board Purpose

Board 1 should provide TICVAI with a centralized **real-time and post-transaction fraud intelligence layer** for commercial transactions.

Its job is to answer:

“How risky is this transaction, and what should TICVAI do about it?”

It should detect suspicious patterns across:

- Online payments
- POS transactions
- Mobile App

- Kiosk
- B2B / Reseller
- Refunds
- Cancellations
- Exchanges
- Chargebacks
- Repeated payment attempts
- High-velocity purchasing
- Unusual transaction behavior

The architecture should be:

Transaction + Customer + Account + Payment Signals + Device + Channel + Ticket/Product + Historical Behavior + Refund History + Chargeback History

↓

Fraud Intelligence Engine

Rules + AI/ML + Anomaly Detection + Velocity Analysis + Behavioral Analysis

↓

Risk Score & Risk Classification

Low → Medium → High → Critical

↓

Governed Decision

Allow / Monitor / Challenge / Hold / Human Review / Recommend Block

↓

Owning TICVAI Module

Payment / Ticketing / Refund / Account / Finance

A critical architecture principle:

Fraud AI detects, scores, explains and recommends. It must not have unrestricted authority to block payments, cancel tickets, suspend accounts or invalidate credentials.

Those actions must follow TICVAI's Fraud Policy + AI Governance + RBAC/PBAC + approval requirements.

Screen 1 — Fraud Intelligence Command Center

Purpose

Provide Fraud, Finance, Operations and authorized management users with one central view of payment and transaction fraud activity.

Header KPIs

- Transactions Today

- Transactions Screened
- Approved Transactions
- Flagged Transactions
- High-Risk Transactions
- Critical-Risk Transactions
- Transactions Held
- Suspected Fraud Value
- Refund Risk Value
- Chargeback Value
- Fraud Rate
- False Positive Rate

Example

Transactions Screened

124,820

Flagged

1,482

High Risk

286

Critical

42

Potential Fraud Exposure

AED 186K

Risk Distribution

Donut:

- Low — 94.2%
- Medium — 4.1%
- High — 1.4%
- Critical — 0.3%

Fraud Trend

Timeline showing:

Transaction Volume

vs.

Flagged Transactions

vs.

Confirmed Fraud

Fraud by Channel

Compare:

- B2C Website
- Mobile App
- POS
- Kiosk
- B2B
- Reseller
- OTA/API
- Call Center

Fraud Categories

Show:

- Payment Abuse
- Velocity Abuse
- Refund Abuse
- Chargeback Risk
- Promotion Abuse
- Account/Payment Mismatch
- Suspicious Device
- Transaction Anomaly
- Other

Alert Feed

Example:

CRITICAL

18 transactions from related accounts using multiple cards within 6 minutes.

HIGH

Account generated AED 8,400 refund requests within 48 hours.

Actions

[View Transactions](#) | [View Alerts](#) | [Review High Risk](#) | [View Refund Risk](#) | [View Chargebacks](#)

Screen 2 — Fraud Strategy, Risk Model & Policy Configuration

Purpose

Configure how TICVAI evaluates transaction fraud and translates detected risk into controlled outcomes.

This screen defines the **fraud strategy**, not individual detailed fraud rules.

Fraud Strategy Profile

Fields:

- Strategy Name
- Tenant
- Venue
- Business Unit
- Channel
- Transaction Type
- Currency
- Effective Dates
- Risk Model
- Risk Threshold Profile
- Governance Profile
- Status

Transaction Types

Configure coverage for:

- Sale
- Reservation
- Payment
- Partial Payment
- Refund
- Partial Refund
- Cancellation
- Exchange
- Upgrade
- Reschedule
- Stored Value
- Wallet Top-Up
- Membership Purchase
- Voucher Purchase
- B2B Transaction

Risk Levels

Example:

Score	Classification
0-29	Low
30-59	Medium

Score	Classification
60–79	High
80–100	Critical

Thresholds must be configurable.

Decision Matrix

Example:

Risk	Default Treatment
Low	Allow
Medium	Allow + Monitor
High	Challenge / Review
Critical	Hold / Escalate

Actual permitted actions remain subject to governance.

Scope Hierarchy

Support:

Platform

- Tenant
- Venue
- Channel
- Transaction Type
- Product

Different tenants may therefore have different fraud tolerance.

Strategy Modes

Support:

- Rules Only
- AI/ML Only where approved
- Hybrid
- Monitoring Only
- Simulation
- Production

For TICVAI, I recommend **Hybrid** as the normal strategy.

Actions

Create Strategy | Clone | Test | Activate | View Versions

Screen 3 — Fraud Rule, Velocity & Threshold Studio

Purpose

Allow authorized administrators to configure deterministic fraud controls and velocity rules.

AI should not replace basic deterministic fraud controls.

Rule Categories

Transaction

Examples:

- Transaction Value
- Transaction Frequency
- Quantity
- Product
- Channel
- Currency

Payment

- Multiple Payment Instruments
- Repeated Declines
- Payment Method Changes
- Billing / Customer Mismatch where data is available
- Payment Token Reuse

Velocity

Examples:

More than 5 purchases within 10 minutes.

More than AED 10,000 purchase value within 1 hour.

More than 3 different payment instruments per account within 30 minutes.

Refund Rules

Examples:

More than 3 refunds within 7 days.

Refund value exceeds AED 5,000 within 30 days.

Rule Builder

Use:

IF

Condition

AND / OR

Condition

THEN

Risk Contribution / Flag / Review Recommendation.

Example:

IF

Transactions > 5

WITHIN 10 minutes

AND

Distinct Cards > 3

THEN

Risk +35

Severity = HIGH

Rule Configuration

- Rule Name
- Category
- Scope
- Condition
- Threshold
- Time Window
- Risk Contribution
- Severity
- Priority
- Effective Date
- Exception
- Status

Rule Conflict Detection

Identify:

- Duplicate Rules
- Contradictory Rules
- Overlapping Rules
- Excessively Broad Rules

Actions

Create Rule | Test Rule | Clone | Disable | Simulate

Screen 4 — Real-Time Transaction Risk Screening

Purpose

Provide the transaction-level decisioning engine used when TICVAI needs fraud assessment during a commercial transaction.

This is one of the most important screens.

Transaction View

Example:

Transaction

TXN-984521

Customer

CUST-38271

Channel

B2C

Amount

AED 4,850

Products

8 × Premium Admission

Risk Score

78 / 100

Risk Level

HIGH

Recommended Treatment

CHALLENGE / REVIEW

Risk Signals

Example:

Signal	Contribution
New Account	+8
High Transaction Value	+12
4 Payment Attempts	+15
3 Different Cards	+20
Device Previously Seen	-8
Normal Booking Window	-4

Signal	Contribution
Unusual Purchase Velocity	+22

Screening Flow

Transaction Request

↓

Collect Available Signals

↓

Deterministic Rules

↓

AI/ML Risk Assessment

↓

Risk Aggregation

↓

Governance Check

↓

Decision

Possible Outcomes

- Allow
- Allow + Monitor
- Step-Up / Challenge
- Hold
- Manual Review
- Recommend Decline
- Block only if explicitly governed

Performance Requirement

Real-time screening must be designed for low latency.

Fraud intelligence should not add unnecessary checkout delay.

Resilience

If Fraud AI is unavailable, TICVAI must have configurable fallback behavior.

For example:

AI unavailable

↓

Deterministic Fraud Rules

↓

Payment Provider Controls

↓

Configured Fallback Policy

Core sales should not fail simply because an AI model is temporarily unavailable unless a tenant's policy explicitly requires fail-closed behavior.

Actions

[View Risk Details](#) | [View Signals](#) | [Challenge](#) | [Send to Review](#) | [View Trace](#)

Screen 5 — Payment Instrument, Device & Transaction Pattern Intelligence

Purpose

Detect suspicious relationships between payment instruments, accounts, devices and transactions. The system should look beyond individual transactions.

Relationship View

Conceptually:

Customer A

- Card Token 1
- Device A
- IP / Network Signal
- Transactions

while:

Customer B

- Same Device A
- Card Token 2
- Similar purchasing pattern

This can indicate coordinated abuse.

Important

TICVAI should normally work with **tokenized/payment-provider references**, not raw card details. Sensitive payment data must remain within applicable PCI/payment architecture.

Signals

Where legally and technically available:

- Payment Token
- Payment Method Type
- Device Identifier
- Account
- Session
- IP/Network Indicators

- Channel
- Country/Region
- Transaction Time
- Transaction Frequency
- Decline History
- Purchase Pattern

Pattern Detection

Examples:

One Device → Many Accounts

One Account → Many Cards

One Card Token → Multiple Accounts

Rapid Transaction Attempts

Repeated Decline → Success Pattern

Unusual Transaction Time

Unusual Purchase Value

Relationship Graph

Visual graph showing:

Account

↔ **Device**

↔ **Payment Token**

↔ **Transaction**

↔ **Refund**

↔ **Ticket**

Risk Cluster

Example:

Cluster #FR-182

Accounts:

12

Devices:

3

Payment Tokens:

8

Transactions:

46

Potential Exposure:

AED 21,400

Actions

[View Relationship](#) | [View Cluster](#) | [View Transaction History](#) | [Flag for Investigation](#)

Screen 6 — Refund, Cancellation & Exchange Abuse Intelligence

Purpose

Detect suspicious post-sale behavior involving refunds, cancellations, exchanges, rescheduling and related services.

Refund KPIs

- Refund Requests
- Refund Value
- High-Risk Refunds
- Refund Rate
- Repeat Refund Customers
- Refund After Usage Attempts
- Exchange Abuse Alerts
- Cancellation Abuse Alerts

Customer Refund Profile

Example:

Customer

CUST-10282

Transactions:

14

Refund Requests:

8

Refund Value:

AED 7,820

Refund Rate:

57%

Risk:

HIGH

Abuse Patterns

Detect patterns such as:

- Repeated Purchase → Refund
- High Refund Frequency
- Refund Shortly Before Visit

- Refund After Usage where business rules prohibit it
- Repeated Exchange / Upgrade / Downgrade
- Repeated Reschedule
- Multiple Refund Methods
- Refund Destination Mismatch where detectable
- Repeated Partial Refund Abuse
- Voucher / Wallet Refund Cycling

Timeline

Example:

Purchase

↓

Exchange

↓

Upgrade

↓

Partial Refund

↓

Reschedule

↓

Refund Request

AI can identify unusual lifecycle patterns.

Important Boundary

Fraud AI can say:

Refund request has 87/100 risk.

The **Refund/Finance/Ticketing policy** determines whether the refund is:

- Allowed
- Held
- Escalated
- Rejected

Actions

[Review Refund](#) | [View Lifecycle](#) | [View Risk Signals](#) | [Escalate](#)

Screen 7 — Chargeback & Payment Dispute Intelligence

Purpose

Use historical and current dispute information to identify chargeback exposure and recurring fraud patterns.

Chargeback KPIs

- Open Chargebacks
- Chargeback Value
- Chargeback Rate
- Fraud-Related Chargebacks
- Merchant Disputes
- Recovered Value
- High-Risk Accounts
- High-Risk Products / Channels

Chargeback Analysis

Break down by:

- Channel
- Product
- Venue
- Payment Method
- Country/Region where allowed
- Customer Segment
- Reseller
- Transaction Value
- Booking Lead Time

Pattern Example

TICVAI detects:

Premium Ticket

-

Same-Day Purchase

-

New Account

-

High Quantity

-

Historical Similar Chargebacks

=

Elevated Chargeback Risk

Pre-Transaction Learning

Historical chargebacks should become a signal for future fraud models where legally and technically permitted.

Chargeback Timeline

Transaction

- Ticket Issued
- Admission / Usage
- Chargeback
- Dispute
- Outcome

Evidence References

Where available:

- Transaction
- Ticket
- Payment Authorization
- Customer
- Usage / Access
- Terms Accepted
- Refund History

This is useful for investigation and downstream dispute handling.

Boundary

This screen provides **chargeback intelligence**.

It should not replace the payment gateway or Finance dispute/accounting processes.

Actions

[View Chargeback](#) | [Link Transaction](#) | [View Evidence](#) | [Send to Investigation](#)

Screen 8 — Fraud Risk Explanation & Decision Trace

Purpose

Explain why TICVAI classified a transaction as risky and provide complete traceability.

This integrates directly with the AI Explainability & Audit module already designed.

Transaction

TXN-984521

Risk Score

78 / 100

Risk

HIGH

Primary Risk Factors

1. Multiple Cards — High
2. Purchase Velocity — High
3. Transaction Value — Medium
4. New Account — Medium
5. Device History — Low Risk Signal

Decision Timeline

13:42:02.110

Transaction Received

↓

13:42:02.145

Fraud Rules Evaluated

↓

13:42:02.191

AI Risk Model Evaluated

↓

13:42:02.205

Risk = 78

↓

13:42:02.211

Governance Decision:

Human Review Required

Trace References

- Transaction ID
- Fraud Assessment ID
- Rule Version
- Model Version
- Provider/Runtime
- Data Cut-Off
- Governance Decision ID
- Approval ID if applicable

- Final Transaction Outcome

Explanation Levels

Business

Transaction was flagged due to unusually high purchase velocity and use of multiple payment instruments.

Fraud Analyst

Detailed signals, thresholds and historical comparisons.

Technical / Audit

Model/rule/version/trace metadata.

Important

Do not expose hidden model chain-of-thought.

Use structured:

Signals + Rules + Evidence + Model Output + Governance Decision

Actions

[View Full Trace](#) | [View Rules](#) | [View Evidence](#) | [Export Audit Record](#)

Screen 9 — Fraud Alert, Review & Decision Workspace

Purpose

Provide authorized fraud analysts with a consolidated workspace for reviewing suspicious transactions before they become full investigation cases.

Board 3 will handle deeper case management.

Review Queue

Prioritize by:

- Critical
- High
- Medium
- Exposure Value
- SLA
- Customer Impact
- Transaction Status

Alert Card

Example:

Alert

FRAUD-ALR-29281

Risk:

Critical — 92

Amount:

AED 12,800

Channel:

B2C

Status:

Held for Review

Review Workspace

Show:

- Transaction Summary
- Customer History
- Payment Signals
- Device Signals
- Related Transactions
- Refund History
- Chargebacks
- Rules Triggered
- AI Risk
- Explanation
- Evidence
- Previous Human Decisions

Analyst Decisions

Depending on permission:

- Confirm Legitimate
- Confirm Suspicious
- Allow
- Continue Monitoring
- Request Verification
- Maintain Hold
- Escalate
- Create Fraud Case
- Recommend Block

Reason Codes

Required for material decisions.

Examples:

- Confirmed Legitimate
- Customer Verified
- Known Fraud Pattern

- Account Compromise Suspected
- Payment Abuse
- Refund Abuse
- Insufficient Evidence
- False Positive
- Other

False Positive Capture

If analyst says:

Legitimate

the outcome should become feedback for model/rule evaluation.

It should **not automatically retrain production models**.

SLA

Example:

Critical:

5 minutes

High:

30 minutes

Medium:

4 hours

Configurable.

Actions

Approve | Maintain Hold | Escalate | Create Case | View Full History

Screen 10 — Fraud Performance, Threshold Simulation & Optimization

Purpose

Measure whether fraud controls are actually effective and allow administrators to test threshold changes before production deployment.

This is critical because:

Blocking more transactions does not automatically mean better fraud prevention.

We must balance fraud prevention against legitimate customer conversion.

Fraud Performance KPIs

- Fraud Detection Rate
- Confirmed Fraud
- False Positive Rate
- False Negative Estimate where measurable
- Approval Rate

- Challenge Rate
- Review Rate
- Hold Rate
- Chargeback Rate
- Prevented Exposure
- Customer Impact

Example

Current Strategy:

Flag Rate

1.7%

Confirmed Fraud

0.31%

False Positive Rate

0.42%

Chargeback Rate

0.28%

Threshold Simulator

Current:

High Risk = 60

Critical = 80

Test Scenario:

High Risk:

65

Critical:

85

Estimated impact:

- Reviews ↓ 18%
- False Positives ↓ 12%
- Potential Fraud Exposure ↑ 4%

Rule Simulation

Example:

New rule:

More than 4 cards within 20 minutes.

Backtest against historical transactions.

Show:

Transactions Matched

2,840

Confirmed Fraud Matched

382

Legitimate Transactions Matched

164

Model Comparison

Compare:

- Current Production Model
- Candidate Model
- Rules Only
- Hybrid

Metrics:

- Detection
- Precision
- Recall where appropriate
- False Positive Rate
- Commercial Impact
- Latency

Segment Performance

Analyze by:

- Venue
- Channel
- Product
- Transaction Type
- Value Band
- Customer Type
- Payment Type

This helps detect whether a rule unfairly or incorrectly impacts a particular legitimate business segment.

Actions

Run Simulation | Backtest Rule | Compare Strategy | Create Change Request

Production changes should follow controlled governance/versioning.

Critical Fraud Architecture

Soft Labs should **not build Fraud AI as one LLM asking whether a transaction looks suspicious.**

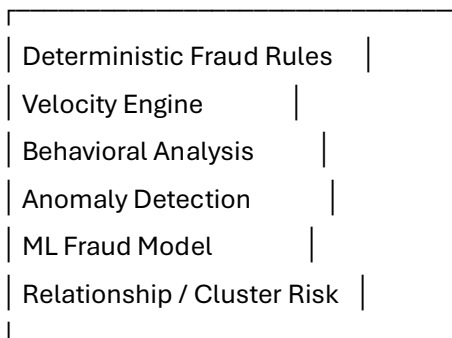
The correct architecture should be hybrid:

TRANSACTION

- |
- |— Transaction Data
- |— Customer / Account
- |— Payment Token / Provider Signals
- |— Device / Session
- |— Channel
- |— Product / Ticket
- |— Historical Transactions
- |— Refund History
- |— Chargeback History



SIGNAL PROCESSING



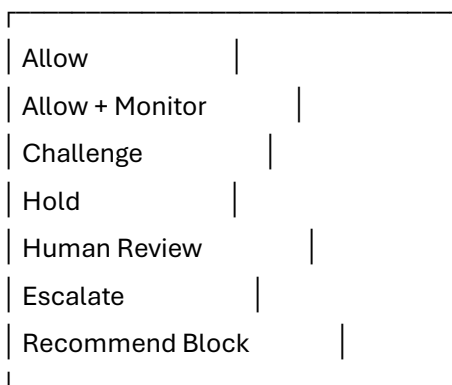
RISK AGGREGATION



FRAUD ASSESSMENT OBJECT



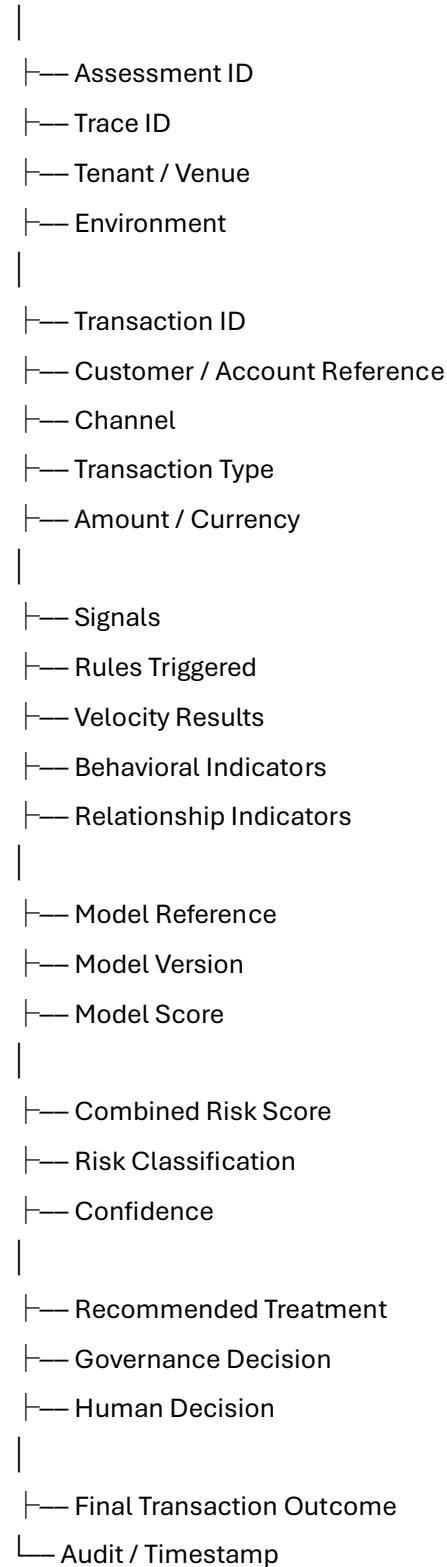
AI GOVERNANCE ENGINE



Fraud Assessment Object

Every fraud evaluation should create a versioned, auditable object.

FraudAssessment



This object becomes the common link between Board 1, AI Governance, Explainability and Board 3 investigation.

Real-Time vs Post-Transaction Detection

Fraud intelligence must support **both**.

Real-Time

Before/during transaction:

Transaction

- Screen
- Score
- Governed Decision
- Continue / Challenge / Hold

Post-Transaction

Some fraud becomes visible only afterward:

Transaction completed

↓

New related activity appears

↓

Pattern changes

↓

Risk increases

↓

Alert generated

For example, ten accounts may initially appear unrelated. Later TICVAI discovers they are associated with the same device/payment pattern.

Therefore fraud screening cannot end when checkout completes.

Risk Score vs Decision

This distinction is critical.

Fraud Engine says:

Risk = 87 / Critical

That is an **assessment**.

Governance determines:

What is TICVAI allowed to do because Risk = 87?

For Tenant A:

Hold + Human Review

For Tenant B:

Challenge customer

For another environment:

Monitor only

So:

Risk Score ≠ Business Action

Rules vs AI

TICVAI should use both.

Deterministic Rule

More than 5 transactions in 10 minutes.

Easy to understand and enforce.

ML / AI

May detect:

This combination of transaction value, purchase timing, account age, device relationship and behavior resembles historical fraud even though no single threshold was violated.

Best Architecture

Rules + ML/AI + Behavioral + Relationship Intelligence

rather than choosing only one.

Payment Gateway Boundary

This is important for Soft Labs.

The payment gateway may already provide:

- Authorization Result
- Decline Reason
- AVS where applicable
- 3DS Result
- Provider Risk Signal
- Payment Method Information
- Tokenized Card Reference

TICVAI should **consume those permitted signals**.

It should not attempt to recreate the payment processor itself.

Conceptually:

TICVAI Fraud Intelligence

-

Payment Provider Fraud Signals

=

Combined Business Risk Context

PCI & Sensitive Payment Data

TICVAI Fraud should avoid unnecessary storage of raw card data.

Use:

- Payment Tokens
- Provider References
- Masked Information
- Approved Derived Signals

rather than full payment credentials.

The architecture should minimize PCI exposure.

Refund Fraud Boundary

Fraud AI says:

Refund Risk = High.

Ticket Service / Refund Policy says:

Is this ticket legally and commercially refundable?

Finance/Payment says:

How will money be returned?

These remain separate responsibilities.

Access Fraud Boundary

Board 1 is **commercial transaction fraud**.

Examples:

- Payment
- Purchase
- Refund
- Chargeback

Board 2 will cover **ticket/account/credential behavioral fraud**, such as:

- Credential Sharing
- Repeated QR Abuse
- Suspicious Scan Behavior
- Account Abuse
- Ticket Transfer Abuse
- Impossible Usage
- Multi-device behavior
- Access-related fraud

This prevents Board 1 from duplicating our Access Control module.

Board 1 → Board 2 → Board 3

The Fraud module should have a clean progression:

Board 1

Is the commercial transaction suspicious?

Payment → Transaction → Refund → Chargeback

↓

Board 2

Is the customer/account/ticket/credential behavior suspicious?

Account → Device → Ticket → Transfer → Scan → Access → Behavior

↓

Board 3

What do we do with the suspicious activity?

Alerts → Evidence → Investigation → Case → Link Analysis → Human Decision → Action → Outcome

False Positives

This deserves special attention.

Suppose TICVAI sees:

AED 8,000 transaction

-

New account

-

10 tickets

This may look suspicious.

But it could simply be:

A father buying tickets for a family group.

Therefore the objective is **not**:

Block every unusual transaction.

It is:

Identify risk accurately while minimizing impact on legitimate customers.

That is why Screen 10 must measure:

Fraud Detection

against

False Positives + Conversion Impact + Customer Impact.

Cold Start

A new TICVAI tenant may have no fraud history.

The system should support:

- Platform Baseline
- Industry/Category Baseline where legally and contractually appropriate
- Deterministic Rules
- Payment Provider Signals
- Tenant-Specific Rules
- Gradual Model Learning
- Reduced Confidence

Tenant data should not automatically be pooled across customers.

Multi-Tenant Isolation

Fraud data is highly sensitive.

Tenant A must never be able to access:

- Tenant B customers
- Fraud cases
- Payment behavior
- Rules
- Risk models
- Alerts

Tenant isolation must be server-side.

AI Governance Integration

The Fraud module should integrate directly with the Governance boards already completed.

Example:

Fraud Risk = Critical

↓

Governance evaluates:

- Tenant
- Capability
- Action
- User
- Transaction Value
- Environment
- Risk
- Autonomy

↓

Decision:

Human Approval Required

↓

Human Oversight

↓

Owning Module executes.

Explainability Integration

Every material fraud decision should connect to:

Trace ID

and show:

Signals → Rules → Model → Risk → Governance → Human Decision → Final Outcome

This will be particularly important when a legitimate customer asks:

Why was my transaction held?

Internally, TICVAI needs an auditable answer.

Board 1 — Final Screen Map

Payment, Transaction, Refund & Chargeback Fraud

- 1. Fraud Intelligence Command Center**
- 2. Fraud Strategy, Risk Model & Policy Configuration**
- 3. Fraud Rule, Velocity & Threshold Studio**
- 4. Real-Time Transaction Risk Screening**
- 5. Payment Instrument, Device & Transaction Pattern Intelligence**
- 6. Refund, Cancellation & Exchange Abuse Intelligence**
- 7. Chargeback & Payment Dispute Intelligence**
- 8. Fraud Risk Explanation & Decision Trace**
- 9. Fraud Alert, Review & Decision Workspace**
- 10. Fraud Performance, Threshold Simulation & Optimization**

So the Board 1 flow is:

CONFIGURE → DETECT → SCORE → EXPLAIN → REVIEW → LEARN → OPTIMIZE

while keeping the most important TICVAI principle:

AI identifies the risk. Governance controls the authority. The owning business module executes the final action.

AI Fraud & Risk Intelligence — Board 2: Ticket, Account, Credential & Behavioral Fraud

Module: AI Fraud & Risk Intelligence

Board: 2 of 3

Screens: 10

Board Type: Ticket Fraud / Account Abuse / Credential Sharing / Access Behavior / Transfer Abuse / Device Intelligence / Behavioral Risk



Board Purpose

Board 1 focused on:

Is the commercial transaction suspicious?

Board 2 focuses on what happens **after** or **around** the purchase:

Is the customer, account, ticket or credential being used in a suspicious or abusive way?

This board should detect fraud involving:

- Ticket sharing
- Credential sharing

- Dynamic QR abuse
- Screenshot / copied credential attempts
- Multiple-device usage
- Suspicious ticket transfers
- Account takeover indicators
- Repeated invalid scans
- Impossible or abnormal usage
- Membership / annual pass abuse
- Wallet / entitlement abuse
- Re-entry abuse
- Group-ticket abuse
- Reseller-related ticket patterns
- Bot / automated account behavior
- Coordinated fraud across multiple accounts
- Suspicious identity/credential relationships

The high-level architecture should be:

Account + Customer + Ticket + Credential + Device + Transfer + Access Scan + Membership + Entitlement + Wallet + Channel + Historical Behavior

↓

Fraud Behavioral Intelligence Engine

Deterministic Rules + Velocity + Relationship Analysis + Behavioral Baseline + Anomaly Detection + AI/ML Risk

↓

Behavioral Risk Score

↓

Low / Medium / High / Critical

↓

Governance Decision

↓

Monitor / Challenge / Review / Recommend Restriction / Escalate

↓

Owning Module

Ticketing / Access Control / Account / Membership / Wallet / B2B

The fundamental rule remains:

Fraud Intelligence detects suspicious behavior. The owning TICVAI module remains responsible for executing operational actions such as invalidating a ticket, restricting an account or changing access rights.

Screen 1 — Ticket & Behavioral Fraud Command Center

Purpose

Provide one central operational view of ticket, account, credential and access-related fraud across TICVAI.

Header KPIs

- Active Tickets Monitored
- Accounts Monitored
- Credentials Monitored
- Behavioral Alerts Today
- High-Risk Accounts
- High-Risk Tickets
- Credential Sharing Alerts
- Suspicious Transfers
- Repeated Scan Abuse
- Account Takeover Alerts
- Estimated Fraud Exposure
- Confirmed Abuse

Risk Distribution

Show:

- Low
- Medium
- High
- Critical

across:

Accounts | Tickets | Credentials | Memberships

Fraud Trend

Timeline:

Behavioral Alerts

vs.

Confirmed Abuse

vs.

False Positives

Fraud by Type

Examples:

Fraud Type	Alerts
Credential Sharing	184
Ticket Transfer Abuse	96
Repeated Scan Abuse	82
Account/Device Anomaly	71
Membership Abuse	54
Re-Entry Abuse	38
Group Ticket Abuse	29
Entitlement Abuse	24

Fraud by Venue / Zone

Heatmap showing where suspicious usage occurs.

Critical Alerts

Example:

CRITICAL

Same annual membership credential detected across three unrelated devices and two entry locations within an impossible time window.

HIGH

Ticket transferred four times within 35 minutes before admission.

Actions

[View Accounts](#) | [View Tickets](#) | [View Credentials](#) | [Review Alerts](#) | [View Fraud Map](#)

Screen 2 — Behavioral Fraud Strategy & Detection Configuration

Purpose

Configure how TICVAI evaluates account, ticket, credential and behavioral risk.

This is the master strategy layer for Board 2.

Strategy Profile

Fields:

- Strategy Name
- Tenant
- Venue
- Fraud Domain
- Channel

- Credential Type
- Customer Type
- Risk Threshold Profile
- Detection Mode
- Governance Profile
- Effective Dates
- Status

Fraud Domains

Configure:

- Ticket Abuse
- Account Abuse
- Credential Sharing
- Transfer Abuse
- Access Abuse
- Membership Abuse
- Entitlement Abuse
- Re-Entry Abuse
- Group Abuse
- Device Abuse

Detection Modes

- Rules Only
- Behavioral
- AI/ML
- Hybrid
- Monitoring Only
- Simulation

Recommended default:

Hybrid

Risk Thresholds

Example:

Score	Level
0–29	Low
30–59	Medium
60–79	High
80–100	Critical

Default Treatment

Risk	Treatment
Low	Allow
Medium	Monitor
High	Review / Challenge
Critical	Escalate / Recommend Restriction

Scope

Allow configuration by:

Tenant → Venue → Ticket Type → Credential → Channel → Customer Type

Actions

Create Strategy | Clone | Test | Activate | View Versions

Screen 3 — Account, Device & Identity Behavior Intelligence

Purpose

Detect suspicious behavior involving customer accounts, sessions, devices and identity relationships.

Account Risk Profile

Example:

Account

CUST-82172

Risk Score:

84 / Critical

Account Age:

12 days

Devices:

7

Tickets:

18

Transfers:

11

Failed Logins:

14

Active Alerts:

4

Behavioral Signals

Where legally and technically available:

- Account Age
- Login Frequency
- Failed Authentication
- Password / Security Changes
- Device Count
- New Device
- Session Behavior
- Geographic / Network Anomaly
- Account Sharing Indicators
- Purchase Behavior
- Transfer Behavior
- Access Behavior

Device Relationship

Example:

Device A

linked to:

Account 1
Account 2
Account 3
Account 4
Account 5

Risk:

HIGH

Account Takeover Indicators

Potential signals:

Known Account

↓

New Device

•

Unusual Login

•

Security Detail Change

•

Ticket Transfer

•

Different Usage Pattern

=

Possible Account Takeover

The system should identify this as a risk signal, not automatically declare that takeover occurred.

Device Velocity

Example:

Device associated with 12 accounts during the previous 60 minutes.

Relationship Graph

Show:

Account ↔ Device ↔ Session ↔ Ticket ↔ Credential ↔ Transaction

Actions

View Account | View Devices | View Sessions | View Relationships | Create Alert

Screen 4 — Ticket & Credential Sharing Detection

Purpose

Detect when tickets, QR codes, digital credentials, RFID/NFC credentials or membership passes appear to be copied or shared improperly.

This should integrate closely with the Dynamic QR & Access Control module already designed.

Credential Risk Profile

Show:

- Ticket ID
- Credential ID
- Credential Type
- Owner
- Device Binding
- Activation Status
- Last Refresh
- Last Scan
- Scan Count
- Devices Seen
- Risk Score

Credential Types

Support:

- Static Barcode
- QR
- Dynamic QR
- Mobile Ticket

- RFID
- NFC
- Wallet Credential
- Membership Credential
- Wristband
- Other supported media

Sharing Indicators

Examples:

Same Credential → Multiple Devices

Same Credential → Impossible Locations

Dynamic QR → Invalid Historical Version

Credential → Multiple Simultaneous Scan Attempts

Membership → Repeated Different Device Activation

Example

Ticket:

TKT-827192

Expected Device:

Device A

Observed:

Device A

Device B

Device C

Time:

within 18 minutes.

Risk:

HIGH — POSSIBLE SHARING

Dynamic QR Integration

Fraud Intelligence should consume signals such as:

- QR Version
- Timestamp
- Device Binding
- Activation
- Refresh Sequence
- Scan Result
- Credential Status

Access Control remains responsible for validating the credential.

Important Boundary

Access Control says:

Is this credential technically valid right now?

Fraud Intelligence says:

Does the overall pattern of using this credential look suspicious?

These are different questions.

Actions

[View Credential](#) | [View Devices](#) | [View Scan History](#) | [Review Sharing Alert](#)

Screen 5 — Access, Scan & Re-Entry Abuse Intelligence

Purpose

Analyze scan and admission behavior to identify suspicious access patterns.

Scan Intelligence

Monitor:

- Valid Scans
- Invalid Scans
- Duplicate Scans
- Rejected Scans
- Re-Entry
- Exit
- Manual Overrides
- Gate
- Device
- Location
- Time

Example

Credential:

TKT-921882

09:12 — Gate A — Entry — Valid

09:16 — Gate B — Entry Attempt — Duplicate

09:18 — Gate D — Entry Attempt — Rejected

09:23 — Gate C — Manual Override Attempt

Risk:

Critical

Abuse Patterns

Detect:

- Repeated Entry Attempts
- Anti-Passback Violations
- Excessive Re-Entry
- Invalid Credential Reuse
- Multiple Gate Attempts
- Repeated Manual Override Requests
- Entry Without Expected Exit Pattern
- Credential Used After Transfer
- Credential Used After Refund/Cancellation
- Abnormal Scan Velocity

Gate Risk Heatmap

Show:

Gate × Time

with fraud/abuse concentration.

Re-Entry Intelligence

Example:

Configured entitlement:

1 Re-Entry

Observed:

4 Attempts

Risk:

High

Override Monitoring

Manual overrides are important fraud signals.

Show:

- Override User
- Credential
- Gate
- Reason
- Time
- Frequency

Repeated overrides involving the same operator/account/ticket can be flagged for investigation.

Actions

[View Scan Timeline](#) | [View Gate](#) | [Review Override](#) | [Send to Investigation](#)

Screen 6 — Ticket Transfer, Resale & Ownership Abuse Intelligence

Purpose

Detect suspicious ticket ownership changes, transfers, resale patterns and transfer networks.

Transfer KPIs

- Transfers Today
- High-Risk Transfers
- Repeated Transfers
- Multi-Hop Transfers
- Last-Minute Transfers
- Reversed Transfers
- Transfer Network Alerts

Ticket Lifecycle

Example:

Purchased by A

↓

Transferred to B

↓

Transferred to C

↓

Transferred to D

↓

Activated on Device E

↓

Scan Attempt from Device F

This may warrant higher risk.

Transfer Risk Signals

Examples:

- Excessive Transfer Count
- Rapid Transfer Sequence
- Transfer Immediately Before Admission
- Repeated Sender Across Many Accounts
- Repeated Recipient Across Many Tickets
- Transfer After Suspicious Payment

- Transfer After Refund Request
- Transfer to Newly Created Account
- Circular Transfer

Transfer Network

Visual:

Account A

→ 12 tickets

→ Accounts B/C/D/E

→ Further transfers

This may indicate unauthorized resale or organized abuse.

Reseller Boundary

Authorized B2B/reseller activity must not automatically be treated as fraud.

The system must understand:

Authorized Distribution

versus

Unexpected Consumer Transfer Pattern

Risk Example

Ticket:

TKT-62912

Transfers:

5

Time:

42 minutes

Accounts:

5

Final activation:

New Device

Risk:

87 / Critical

Actions

[View Transfer Chain](#) | [View Accounts](#) | [Review Ownership](#) | [Create Fraud Alert](#)

Screen 7 — Membership, Pass, Wallet & Entitlement Abuse Intelligence

Purpose

Detect suspicious usage involving reusable products and entitlements.

Coverage

- Annual Pass
- Membership
- Subscription
- Stored Value
- Wallet
- Loyalty
- Fast Pass
- Ride Entitlements
- F&B Entitlements
- Retail Entitlements
- Packages
- Benefits

Membership Risk

Example:

Annual Pass

PASS-82718

Owner:

Customer A

Visits this month:

18

Devices:

5

Credential Sharing Alerts:

3

Identity mismatch events:

2

Risk:

HIGH

Abuse Patterns

Examples:

Membership Sharing

Benefit Farming

Repeated Guest Discount Abuse

Stored Value Cycling

Unusual Wallet Redemption

Excessive Entitlement Consumption

Pass Usage Beyond Expected Pattern

Fast Pass Sharing

Entitlement Consumption

Show:

Configured Entitlement

vs.

Actual Consumption

vs.

Expected Behavioral Range

Example

Gold Pass:

Ride entitlement:

Maximum 1 use per ride/day.

Observed:

6 attempts at same ride.

Access Control may already reject attempts.

Fraud Intelligence detects:

This behavior forms a recurring abuse pattern.

Wallet Boundary

Wallet owns:

- Balance
- Credits
- Debit
- Redemption

Fraud AI evaluates:

- Behavioral Risk
- Abnormal Usage
- Relationships
- Abuse Pattern

Actions

[View Membership](#) | [View Wallet Behavior](#) | [View Entitlements](#) | [Review Risk](#)

Screen 8 — Behavioral Anomaly & Relationship Intelligence

Purpose

Detect fraud patterns that cannot be identified by looking at one account or ticket individually.

This is where Board 2 becomes significantly more powerful.

Relationship Graph

Nodes:

- Customer
- Account
- Device
- Ticket
- Credential
- Payment Token Reference
- Membership
- Wallet
- Transfer
- Gate
- Scan
- Transaction

Example Cluster

Cluster FR-2982

Accounts:

24

Devices:

5

Tickets:

86

Transfers:

62

Suspicious Scans:

31

Estimated Exposure:

AED 48,600

Pattern

AI detects:

Multiple accounts appear unrelated individually but share devices, ticket-transfer relationships and similar access behavior.

Relationship Strength

Show:

Strong

Same device + same credential

Medium

Repeated transfers

Weak

Similar behavior only

Anomaly Types

- Account Cluster
- Device Cluster
- Credential Cluster
- Transfer Network
- Scan Network
- Membership Sharing Network
- Coordinated Abuse
- Bot-like Activity

Behavioral Baseline

Compare:

Expected Behavior

vs.

Observed Behavior

Example:

Normal family account:

1–3 devices

Observed:

17 devices

Important

Unusual behavior does not automatically equal fraud.

Display:

Anomaly Detected

rather than:

Fraud Confirmed

until evidence/human decision supports confirmation.

Actions

[Explore Graph](#) | [Expand Cluster](#) | [View Evidence](#) | [Send to Investigation](#)

Screen 9 — Behavioral Fraud Alert & Review Workspace

Purpose

Provide fraud analysts with one workspace to review suspicious ticket/account/credential behavior before escalating it into a formal fraud case.

Alert Queue

Prioritize:

- Critical
- High
- Exposure
- Active Usage
- Upcoming Visit
- SLA
- Number of Related Entities

Alert Example

ALR-827182

Type:

Credential Sharing

Risk:

91 / Critical

Ticket:

TKT-72918

Accounts:

3

Devices:

4

Scan Attempts:

7

Exposure:

AED 1,850

Review Workspace

Display:

- Account Summary
- Customer Summary
- Ticket
- Credential
- Device History
- Transfer History
- Scan Timeline
- Membership
- Related Transactions
- Relationship Graph
- Triggered Rules
- AI Risk
- Explanation
- Previous Alerts

Analyst Actions

Subject to permissions:

- Confirm Legitimate
- Confirm Suspicious
- Continue Monitoring
- Request Verification
- Recommend Ticket Restriction
- Recommend Account Restriction
- Recommend Credential Invalidation
- Escalate
- Create Fraud Case

Important

Fraud analyst should not necessarily directly invalidate the credential from this screen.

Example:

Recommend Credential Invalidation

↓

Governance

↓

Access Control

↓

Invalidate Credential

Reason Codes

Examples:

- Credential Sharing

- Account Sharing
- Transfer Abuse
- Membership Abuse
- Re-Entry Abuse
- Account Takeover Suspected
- Coordinated Fraud
- False Positive
- Insufficient Evidence

Actions

Mark Legitimate | Confirm Suspicious | Escalate | Create Case

Screen 10 — Behavioral Fraud Performance, Simulation & Optimization

Purpose

Measure how well Board 2 detects abuse and allow rule/model changes to be tested before production.

Performance KPIs

- Behavioral Alerts
- Confirmed Abuse
- False Positive Rate
- High-Risk Tickets
- High-Risk Accounts
- Credential Sharing Detection
- Transfer Abuse Detection
- Access Abuse Detection
- Prevented Exposure
- Customer Impact

Detection Performance

Compare:

Rules

vs.

Behavioral Model

vs.

Hybrid

Example

Strategy	Detection	False Positive	Alerts
Rules	71%	0.8%	1,820
Behavioral	82%	1.2%	2,210
Hybrid	89%	0.7%	1,940

Only display metrics where a valid labeled outcome supports them.

Rule Simulation

Example:

Current rule:

More than 3 devices per credential within 24 hours.

Proposed:

More than 4 devices within 12 hours.

Backtest:

- Alerts ↓ 18%
- Confirmed Abuse Detection ↓ 2%
- False Positives ↓ 23%

Threshold Simulation

Current:

High ≥ 60

Critical ≥ 80

Proposed:

High ≥ 65

Critical ≥ 85

Show estimated effect.

Performance by Fraud Type

- Credential Sharing
- Transfer Abuse
- Re-Entry
- Membership
- Account
- Device
- Entitlement

Performance by Venue / Product

Identify areas with:

- Excessive alerts

- Poor detection
- High false positives
- Emerging patterns

Optimization Recommendation

Example:

Credential-sharing rule generates high false positives for family ticket products. Review family-account device allowance.

Actions

Run Simulation | Backtest Rule | Compare Models | Create Change Request

Critical Board 2 Architecture

Soft Labs should treat this as a **cross-module behavioral intelligence service**.

ACCOUNT / CUSTOMER

+

TICKET / CREDENTIAL

+

DEVICE / SESSION

+

TRANSFER / OWNERSHIP

+

ACCESS / SCANS

+

MEMBERSHIP / WALLET

+

ENTITLEMENTS

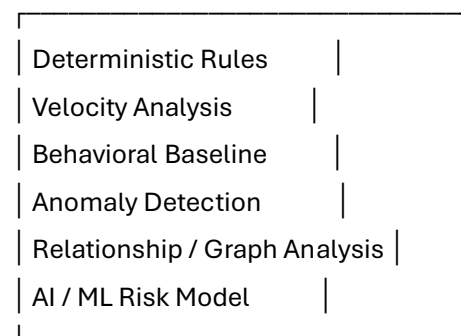
+

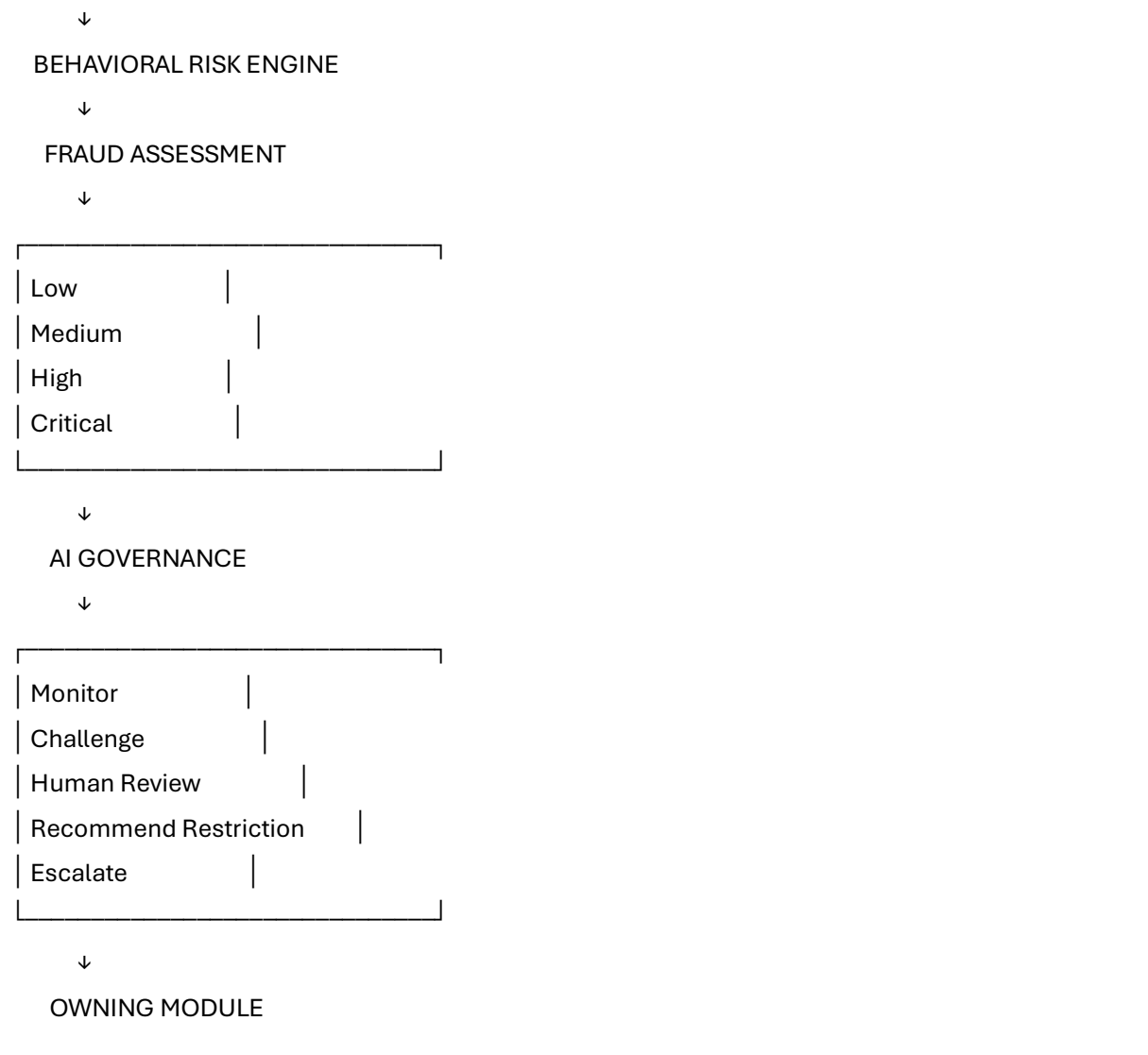
TRANSACTION CONTEXT

↓

BEHAVIORAL SIGNAL LAYER

↓





One Unified Fraud Identity Graph

I strongly recommend Soft Labs does **not** build separate disconnected fraud records for accounts, tickets, devices and credentials.

Create a relationship graph conceptually like:

CUSTOMER

|

ACCOUNT

|

|— DEVICE

|

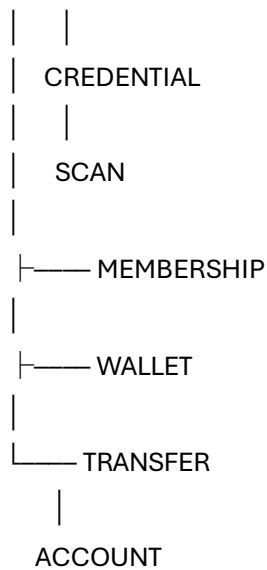
|— SESSION

|

|— PAYMENT TOKEN

|

|— TICKET



This allows TICVAI to detect fraud that would otherwise be invisible.

For example:

10 different accounts

may appear normal independently.

But if they all use:

the same two devices + same payment token + repeatedly transfer tickets between themselves, the relationship becomes suspicious.

Important — Identity Graph ≠ Customer Master

CRM remains owner of the customer master.

The Fraud Identity Graph is a **risk relationship layer** referencing existing entities.

It should not create another CRM database.

Access Control Boundary

This is particularly important because we already designed a very comprehensive Access Control module.

Access Control asks:

Is this credential valid according to access rules?

Example:

QR valid?

Correct time?

Correct gate?

Already used?

Anti-passback allowed?

Fraud Intelligence asks:

Does the behavior around this credential look suspicious?

Example:

Credential is technically valid but has appeared on **five different devices** over two days.

Therefore:

Validation ≠ Fraud Detection.

Dynamic QR Boundary

Dynamic QR already handles:

- Refresh
- OTP / Timestamp
- Device Binding
- Screenshot Protection
- Cryptographic Validation
- Activation
- Invalidation

Fraud AI should **consume the resulting security events**.

It should not rebuild Dynamic QR.

Example:

Dynamic QR

↓

Invalid historical QR attempts

↓

Repeated across multiple devices

↓

Fraud Intelligence

↓

Pattern detected

Ticket Transfer Boundary

Ticketing owns:

- Transfer Eligibility
- Transfer Workflow
- New Ownership
- Ticket Re-Issue

Fraud Intelligence analyzes:

- Frequency
- Velocity
- Network
- Relationship
- Timing
- Historical behavior

So a transfer may be **allowed by policy** but still **suspicious behaviorally**.

Fraud Risk Should Follow the Entity

This is important architecturally.

Risk should not exist only at transaction level.

TICVAI should support risk for:

Transaction

Account

Customer

Ticket

Credential

Device

Membership

Wallet

Fraud Cluster

Example:

Account Risk 82 HIGH

Ticket Risk 91 CRITICAL

Device Risk 76 HIGH

Credential Risk 94 CRITICAL

These can influence each other without simply copying one score to every entity.

Temporal / Impossible Usage Detection

Time relationships are very useful.

Example:

15:02

Credential used at Venue A.

15:07

Same credential appears at Venue B, physically impossible to reach in five minutes.

The system should identify:

Impossible Usage Pattern

This does not necessarily mean automatic blocking.

It creates a strong risk signal.

Family & Group Ticket Protection

Soft Labs must avoid aggressive rules that create false positives for legitimate families/groups.

For example:

One parent may legitimately:

- Hold several children's tickets
- Use one mobile device
- Transfer tickets to family members
- Purchase 10 tickets
- Manage multiple memberships

Therefore fraud rules need **product and relationship context**.

A family account behaving like a family should not be treated the same as 10 unrelated consumer accounts sharing one credential.

Authorized B2B / Reseller Protection

The same applies to B2B.

A reseller may legitimately:

- Issue hundreds of tickets
- Manage multiple customers
- Use centralized devices
- Generate high transaction volume

The fraud engine must know:

Authorized business behavior

versus

Consumer behavior.

Otherwise TICVAI will generate huge false positives.

Behavioral Baseline

TICVAI should learn or calculate expected patterns at appropriate levels.

Examples:

Normal Annual Pass

2 devices

3 visits/month

Normal Family Account

1–4 devices

multiple children's tickets

Normal Reseller

high ticket volume

centralized business account

Then compare:

Expected Behavior

vs.

Observed Behavior.

This is more powerful than fixed rules alone.

AI/ML Role

Again, an LLM should **not** be the primary fraud detector.

Use:

Rules + Statistical Analysis + ML + Anomaly Detection + Graph Intelligence

LLMs may help with:

- Analyst explanations
- Investigation summaries
- Natural-language queries
- Case summaries
- Pattern descriptions

But the real-time fraud decision should be based on structured fraud intelligence.

Feedback Loop

Analyst outcomes should improve future detection:

Alert

↓

Human Review

↓

Legitimate / Suspicious / Confirmed Fraud

↓

Outcome Dataset

↓

Performance Evaluation

↓

Rule / Model Improvement Candidate

↓

Backtest

↓

Approval

↓

Deployment

Again:

Do not automatically retrain and deploy production fraud models based on every analyst decision.

Board 1 + Board 2 Relationship

The two boards should share intelligence.

Example:

Board 1 detects:

Account made suspicious payment.

Risk:

78 High

↓

Ticket is issued.

↓

Board 2 detects:

Ticket transferred three times and used across two devices.

Risk increases.

↓

Relationship graph now shows:

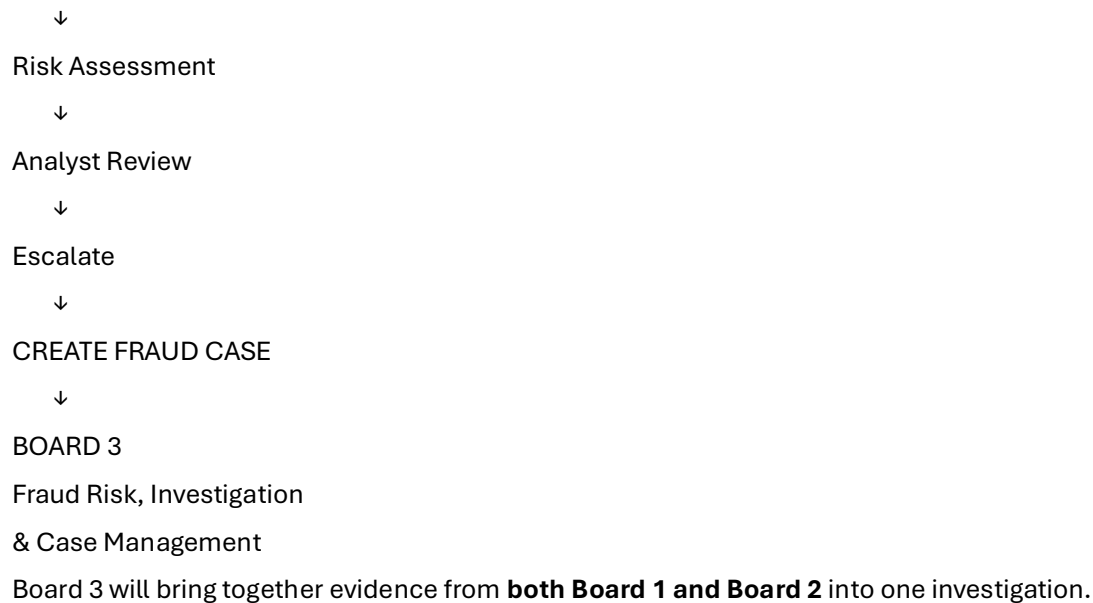
Payment Risk + Transfer Risk + Device Risk + Access Risk

This is much stronger than treating each event independently.

Board 2 → Board 3

When Board 2 identifies something serious:

Behavioral Alert



Board 2 — Final Screen Map

Ticket, Account, Credential & Behavioral Fraud

1. Ticket & Behavioral Fraud Command Center
2. Behavioral Fraud Strategy & Detection Configuration
3. Account, Device & Identity Behavior Intelligence
4. Ticket & Credential Sharing Detection
5. Access, Scan & Re-Entry Abuse Intelligence
6. Ticket Transfer, Resale & Ownership Abuse Intelligence
7. Membership, Pass, Wallet & Entitlement Abuse Intelligence
8. Behavioral Anomaly & Relationship Intelligence
9. Behavioral Fraud Alert & Review Workspace
10. Behavioral Fraud Performance, Simulation & Optimization

So Board 2 follows:

MONITOR → CONNECT → DETECT → SCORE → EXPLAIN → REVIEW → ESCALATE → LEARN

And the distinction across the Fraud module is now very clean:

Board 1 — Money:

Payment, transaction, refund and chargeback fraud.

Board 2 — Behavior:

Account, device, ticket, credential, transfer, access, membership and entitlement abuse.

Board 3 — Investigation:

Bring all evidence together, investigate the fraud network, manage the case, decide the outcome and coordinate governed action

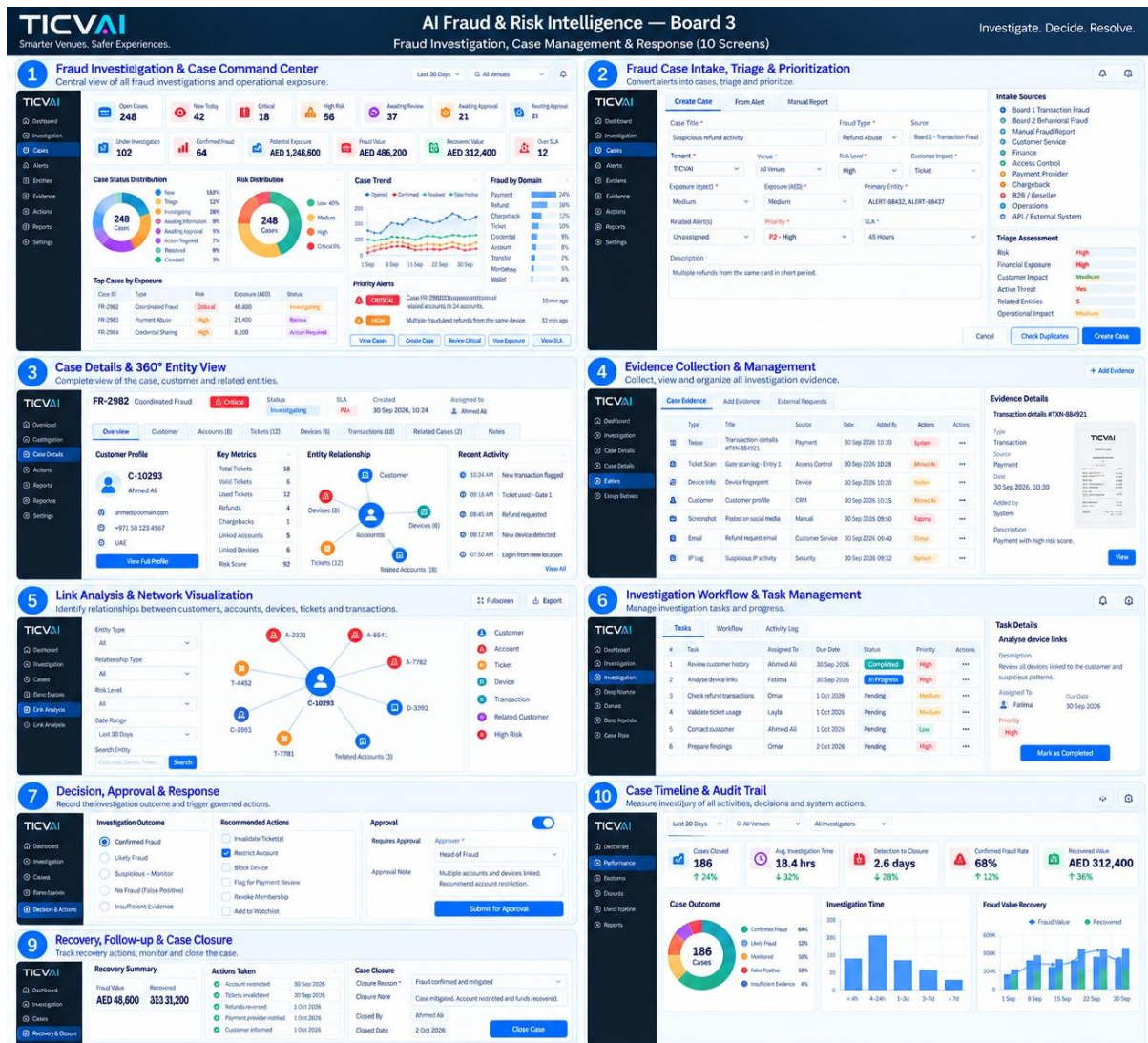
AI Fraud & Risk Intelligence — Board 3: Fraud Investigation, Case Management & Response

Module: AI Fraud & Risk Intelligence

Board: 3 of 3

Screens: 10

Board Type: Fraud Investigation / Case Management / Evidence / Link Analysis / Human Decision / Response / Recovery



Board Purpose

Boards 1 and 2 detect suspicious activity:

Board 1 — Money: payment, transaction, refund and chargeback fraud.

Board 2 — Behavior: account, device, ticket, credential, transfer, access, membership and entitlement abuse.

Board 3 answers the next question:

“Now that TICVAI has detected suspicious activity, how do we investigate it, connect the evidence, make a controlled decision and coordinate the required response?”

The board should provide a complete investigation workflow:

Fraud Alert / Risk Assessment

- Triage
- Create / Link Case
- Collect Evidence
- Connect Related Entities
- Investigate
- Human Decision
- Governance / Approval
- Owning Module Action
- Recovery / Resolution
- Case Closure & Learning

A fundamental architecture principle:

Board 3 manages the investigation and decision process. It should not bypass the business modules that own the actual ticket, account, payment, refund, access or wallet action.

Screen 1 — Fraud Investigation & Case Command Center

Purpose

Provide Fraud Managers and authorized investigators with a central view of all fraud investigations and operational exposure.

Header KPIs

- Open Cases
- New Cases Today
- Critical Cases
- High-Risk Cases
- Cases Awaiting Review
- Cases Awaiting Approval
- Cases Under Investigation
- Confirmed Fraud Cases
- Potential Exposure
- Confirmed Fraud Value
- Prevented / Recovered Value
- Cases Over SLA

Case Status Distribution

Show:

- New
- Triage
- Investigating
- Awaiting Information

- Awaiting Approval
- Action Required
- Monitoring
- Resolved
- Closed

Risk Distribution

Low | Medium | High | Critical

Case Trend

Show:

Cases Opened

vs.

Confirmed Fraud

vs.

Resolved

vs.

False Positive / Legitimate

Fraud Domain

Break down:

- Payment
- Refund
- Chargeback
- Ticket
- Credential
- Account
- Transfer
- Access
- Membership
- Wallet
- Coordinated Fraud

Exposure by Case

Example:

Case	Type	Risk	Exposure	Status
FR-2982	Coordinated Fraud	Critical	AED 48,600	Investigating
FR-2983	Payment Abuse	High	AED 21,400	Review

Case	Type	Risk	Exposure	Status
FR-2984	Credential Sharing	High	AED 8,200	Action Required

Priority Alerts

Example:

CRITICAL

Case FR-2982 expanded from 8 related accounts to 24 accounts.

Actions

[View Cases](#) | [Create Case](#) | [Review Critical](#) | [View Exposure](#) | [View SLA](#)

Screen 2 — Fraud Case Intake, Triage & Prioritization

Purpose

Control how alerts from Boards 1 and 2 become formal fraud investigation cases.

Not every fraud alert should automatically create a case.

Intake Sources

Cases can originate from:

- Board 1 Transaction Fraud
- Board 2 Behavioral Fraud
- Manual Fraud Report
- Customer Service
- Finance
- Access Control
- Payment Provider
- Chargeback
- B2B / Reseller
- Operations
- API / External System

Case Creation

Fields:

- Case ID
- Case Title
- Fraud Type
- Source
- Tenant
- Venue
- Risk Level

- Exposure
- Customer Impact
- Related Alert(s)
- Primary Entity
- Investigator
- Priority
- SLA
- Created Date
- Status

Triage

Evaluate:

Risk

-

Financial Exposure

-

Customer Impact

-

Active Threat

-

Number of Related Entities

-

Operational Impact

Priority

Example:

P1 — Critical

Active fraud / high exposure / immediate action.

P2 — High

Significant suspected fraud.

P3 — Medium

Investigation required.

P4 — Low

Monitor / low impact.

Duplicate Case Detection

TICVAI should detect if a new alert is related to an existing case.

Example:

New alert:

Credential Sharing

Related to:

Case FR-2982

Relationship:

Same Device + Same Account Cluster.

Recommendation:

LINK TO EXISTING CASE

rather than create another disconnected case.

Case Assignment

Assign by:

- Fraud Type
- Venue
- Risk
- Investigator Skill
- Workload
- Region
- SLA

Actions

Create Case | Link Existing Case | Assign | Change Priority | Start Investigation

Screen 3 — Fraud Investigation Workspace

Purpose

Provide the investigator with one complete workspace containing all information required to investigate the case.

This should be the primary working screen for a fraud analyst.

Case Header

Example:

Case

FR-2982

Type

Coordinated Ticket & Payment Fraud

Risk**CRITICAL — 94****Exposure**

AED 48,600

Status

Investigating

Owner

Fraud Analyst A

Case Summary

Show:

- Original Alert
- Fraud Type
- Detection Time
- Risk
- Exposure
- Affected Venue
- Customer Impact
- Related Accounts
- Tickets
- Transactions
- Devices
- Credentials
- Memberships

Investigation Tabs

- Overview
- Alerts
- Accounts
- Transactions
- Tickets
- Devices
- Credentials
- Access
- Transfers
- Refunds
- Chargebacks
- Evidence
- Notes

- Decisions
- Actions

AI Investigation Summary

Example:

Case contains 24 accounts connected through five devices, eight payment-token references and repeated ticket transfers. Twelve tickets generated suspicious access attempts after transfer.

This should be generated from structured evidence.

Investigation Checklist

Example:

- Review original alerts ✓
- Review transactions ✓
- Validate payment relationships ✓
- Review ticket ownership ✓
- Review device graph
- Review access activity
- Review refunds
- Determine exposure
- Record conclusion
- Submit decision

Investigator Notes

Support:

- Notes
- Internal Comments
- Attachments
- Tags
- Mention / Assignment
- Timestamp
- Author

Actions

Add Evidence | Add Note | Expand Investigation | Request Information | Submit Decision

Screen 4 — Fraud Relationship & Network Investigation

Purpose

Provide investigators with a visual graph showing relationships across all fraud entities.

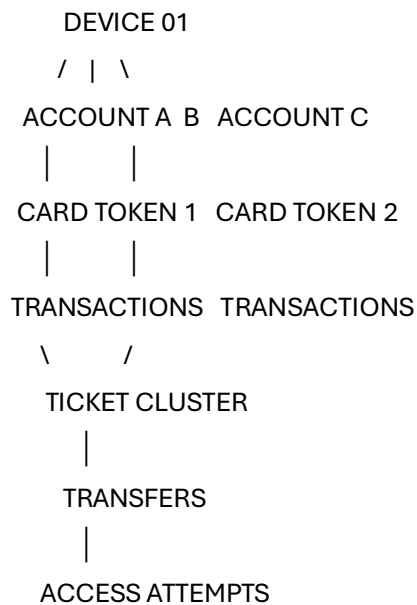
This is one of the most important screens in Board 3.

Graph Entities

Support nodes for:

- Customer
- Account
- Device
- Session
- Payment Token Reference
- Transaction
- Ticket
- Credential
- Membership
- Wallet
- Transfer
- Refund
- Chargeback
- Gate
- Scan
- Reseller

Example



Relationship Types

Examples:

- Same Device
- Same Payment Token
- Same Account
- Ticket Ownership

- Ticket Transfer
- Same Credential
- Same Scan Pattern
- Same Network Signal
- Same Membership
- Same Wallet
- Same Reseller

Relationship Strength

Strong

Direct technical relationship.

Medium

Repeated behavioral relationship.

Weak

Statistical / behavioral similarity.

Cluster Intelligence

Example:

Cluster CL-882

Accounts:

24

Devices:

5

Payment References:

8

Tickets:

86

Transfers:

62

Suspicious Scans:

31

Exposure:

AED 48,600

Expand Network

Investigator can expand:

1 Hop

2 Hops

3 Hops

with permission and performance limits.

Important

Graph association must not automatically mean:

All connected entities are fraudulent.

It means:

A relationship exists and should be evaluated.

Actions

Expand Node | View Entity | Add to Case | Mark Relevant | Export Graph

Screen 5 — Evidence, Timeline & Investigation Audit

Purpose

Create a complete chronological and evidentiary view of the suspected fraud.

Master Timeline

Example:

12 Sep — 10:02

Account A created.

↓

12 Sep — 10:14

AED 8,200 purchase.

↓

12 Sep — 10:17

Three ticket transfers.

↓

12 Sep — 10:22

New device activated.

↓

12 Sep — 10:31

First access attempt.

↓

12 Sep — 10:34

Duplicate credential attempt.

↓

12 Sep — 11:02

Refund requested.

↓

13 Sep

Chargeback received.

Evidence Categories

- Transaction Evidence
- Payment Provider Evidence
- Ticket Evidence
- Credential Evidence
- Device Evidence
- Access Evidence
- Transfer Evidence
- Refund Evidence
- Chargeback Evidence
- Customer Communication
- Analyst Evidence
- External Evidence where approved

Evidence Record

Each item should include:

- Evidence ID
- Type
- Source
- Related Entity
- Timestamp
- Captured By
- Data Reference
- Integrity Reference
- Case Relevance
- Notes

Evidence Integrity

Historical evidence must not be silently overwritten.

If source data changes:

Original Evidence Snapshot

must remain traceable.

Audit

Track:

- Who viewed
- Who added
- Who modified notes
- Who changed case status
- Who made decision
- Who approved action

Actions

[Add Evidence](#) | [View Source](#) | [Add Note](#) | [Download Evidence Package](#) | [View Audit](#)

Screen 6 — Fraud Risk, Exposure & Impact Assessment

Purpose

Determine the real business impact of the suspected fraud rather than relying only on a risk score.

Case Risk

Example:

Fraud Probability / Risk Score

94 / Critical

Financial Exposure

Break down:

- Transaction Value
- Refund Exposure
- Chargeback Exposure
- Ticket Value
- Wallet / Stored Value
- Membership Value
- Potential Future Exposure

Example

Transaction:

AED 28,000

Tickets:

AED 14,400

Potential Refund:

AED 8,200

Wallet:

AED 2,000

Estimated Total Exposure

AED 52,600

Avoid double-counting overlapping exposure.

Operational Impact

Assess:

- Tickets affected
- Customers affected
- Venues affected
- Credentials affected
- Access attempts
- Reseller impact
- Operational disruption

Customer Impact

Show:

- Legitimate Customers Potentially Affected
- Accounts Under Review
- Tickets Held
- Pending Visits
- Customer Verification Required

Risk Dimensions

Example:

Dimension	Level
Financial	Critical
Access	High
Customer	Medium
Operational	High
Reputation	Medium

Exposure Scenarios

Current Exposure

vs.

Potential Exposure if No Action

Example:

Current:

AED 48,600

Potential 48-hour exposure:

AED 92,000

Clearly label projections as estimates.

Actions

Update Assessment | Run Exposure Scenario | View Impact | Submit Risk Review

Screen 7 — Investigation Decision & Fraud Disposition

Purpose

Allow the authorized investigator to record the formal conclusion of the investigation.

Detection and investigation should end with a structured disposition.

Possible Outcomes

- Confirmed Fraud
- Likely Fraud
- Suspicious / Continue Monitoring
- Legitimate
- False Positive
- Insufficient Evidence
- Duplicate / Related Case
- Policy Abuse but Not Fraud
- Other Configured Outcome

Decision Summary

Example:

Case

FR-2982

Investigator Conclusion

Confirmed Fraud

Confidence:

High

Primary Fraud Type

Coordinated Ticket Resale & Credential Abuse

Supporting Evidence

- 24 related accounts
- 5 shared devices
- 62 ticket transfers
- 31 suspicious scans

- 8 payment references
- repeated refund pattern

Decision Reason

Require structured reason codes plus investigator comments.

Recommended Responses

Potential recommendations:

- Restrict Account
- Invalidate Credential
- Suspend Ticket
- Hold Refund
- Review Payment
- Restrict Transfer
- Monitor Device
- Restrict Membership
- Escalate Reseller
- No Action

Important

The investigator's conclusion:

Confirmed Fraud

does not automatically mean:

Block everything.

The response must still pass governance and owning-module policy.

Actions

Save Decision | Request Peer Review | Submit for Approval | Close as Legitimate

Screen 8 — Fraud Response, Approval & Action Orchestration

Purpose

Translate investigation decisions into controlled actions across TICVAI.

This screen should orchestrate actions — not directly replace the owning modules.

Recommended Action Package

Example:

Case FR-2982

Recommended:

1. Suspend 8 accounts
2. Invalidate 12 credentials

3. Hold 3 refund requests
4. Restrict 24 ticket transfers
5. Monitor 5 devices
6. Escalate reseller account

Action Matrix

Action	Owner	Risk	Approval	Status
Suspend Account	Account	High	Required	Pending
Invalidate QR	Access Control	High	Required	Pending
Hold Refund	Finance/Refund	High	Required	Approved
Restrict Transfer	Ticketing	Medium	Required	Pending
Monitor Device	Fraud	Low	No	Active

Governance Flow

Investigator Recommendation

↓

AI Governance

↓

RBAC/PBAC

↓

Human Approval where required

↓

Execution Authorization

↓

Owning Module

↓

Execution Result

Revalidation

Immediately before execution, verify:

- Case still valid
- Ticket/account status
- Governance policy
- Approval validity
- User permission
- Action version

Partial Execution

A case may contain ten actions.

Some may:

Succeed

while others:

Fail / Pending / Rejected

Track individually.

Emergency Action

For explicitly governed high-risk situations, support controlled emergency response such as:

Temporary Credential Hold

with:

- permission
- reason
- expiry
- audit
- retrospective review

Actions

Submit Approval | Execute Approved | Retry Failed | View Governance Trace

Screen 9 — Recovery, Monitoring & Case Resolution

Purpose

Manage what happens after fraud action has been taken and determine when a case can be resolved.

Post-Action Monitoring

Track:

- New Transactions
- New Accounts
- Device Activity
- Ticket Activity
- Scan Attempts
- Transfers
- Refund Attempts
- Chargebacks
- Related Alerts

Example

Case:

FR-2982

Accounts restricted:

8

Credentials invalidated:

12

New Activity

New related account detected:

1

New device:

1

New transfer attempt:

3

Recommendation:

KEEP CASE OPEN

Recovery Tracking

Where applicable:

- Fraud Prevented
- Refund Prevented
- Chargeback Recovered
- Ticket Value Protected
- Wallet Value Protected
- Financial Recovery

Customer Remediation

For legitimate customers affected by fraud controls, track actions such as:

- Customer Verified
- Ticket Restored
- Account Restored
- Credential Reissued
- Refund Released
- Apology / Customer Service Follow-Up

The actual customer-service action remains with the relevant operational module/team.

Resolution Criteria

Configurable examples:

- Required Actions Completed
- No Active Critical Alerts
- Monitoring Period Completed
- Financial Outcome Recorded

- Customer Impact Resolved
- Investigation Decision Complete

Case Closure

Statuses:

- Resolved — Fraud Confirmed
- Resolved — Legitimate
- Resolved — Loss
- Resolved — Recovered
- Closed — No Further Action
- Closed — Duplicate
- Monitoring

Actions

Continue Monitoring | Resolve Case | Reopen | Close Case

Screen 10 — Fraud Case Performance, Learning & Management Reporting

Purpose

Measure the effectiveness of TICVAI's entire fraud investigation operation and feed verified outcomes back into future fraud detection.

Investigation KPIs

- Cases Created
- Cases Closed
- Confirmed Fraud
- Legitimate / False Positive
- Average Investigation Time
- Average Time to Decision
- SLA Compliance
- Open Exposure
- Prevented Fraud
- Recovered Value
- Customer Impact
- Reopened Cases

Funnel

Alerts

10,000

↓

Reviewed

2,400

↓

Cases

620

↓

Confirmed Fraud

280

↓

Actions

240

↓

Resolved

225

This is important because an alert is not the same as confirmed fraud.

Performance by Fraud Type

Compare:

- Payment
- Refund
- Chargeback
- Credential
- Transfer
- Access
- Membership
- Account
- Coordinated Fraud

Investigator / Team Performance

Authorized managers can review:

- Assigned Cases
- Completed Cases
- SLA
- Backlog
- Escalations
- Review Quality

Avoid simplistic employee ranking based only on number of cases closed.

Detection Feedback

Case outcomes should feed back into:

Board 1

Transaction Fraud Performance
and

Board 2

Behavioral Fraud Performance.

Example:

Rule:

Multiple Devices

Alerts:

1,200

Cases:

180

Confirmed Fraud:

42

False Positives:

96

This may indicate the rule needs refinement.

Emerging Fraud Patterns

Example:

Increased credential-sharing cases associated with annual passes during weekends.

New transfer cluster detected across several newly created accounts.

These should be evidence-grounded summaries.

Management Reporting

Support:

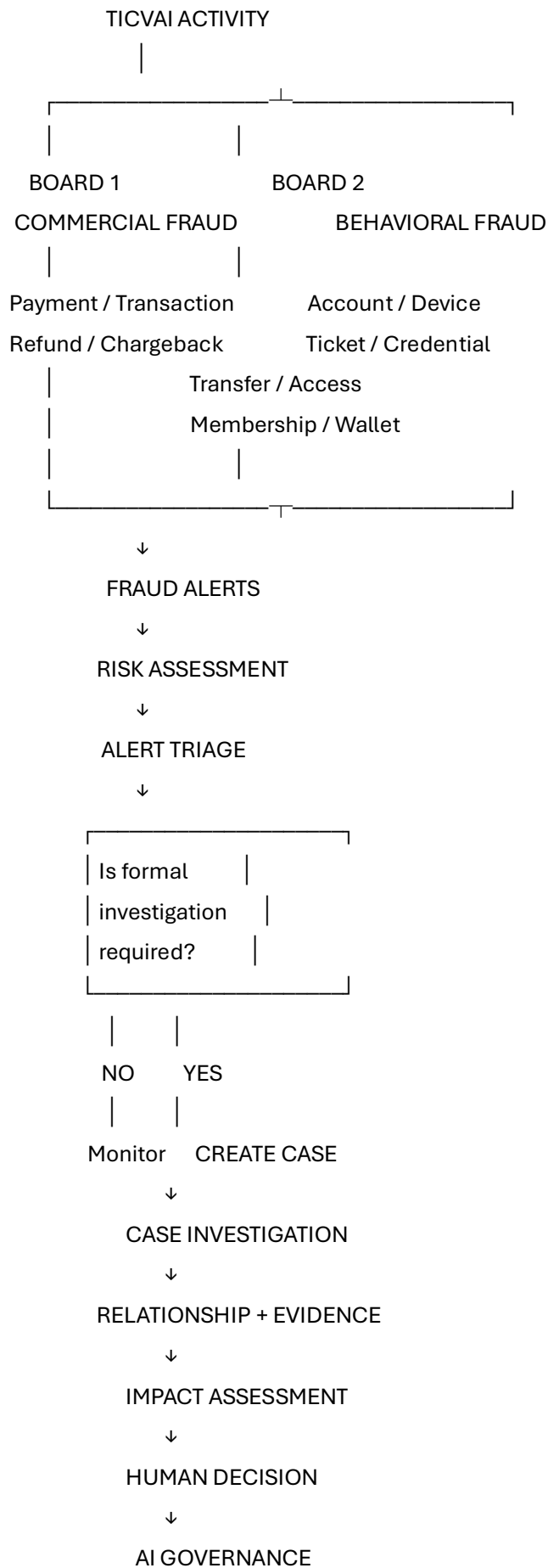
- Daily Fraud Report
- Weekly Fraud Summary
- Monthly Fraud Review
- Exposure Report
- Fraud Loss Report
- Prevention Report
- Case SLA Report
- Fraud Type Report

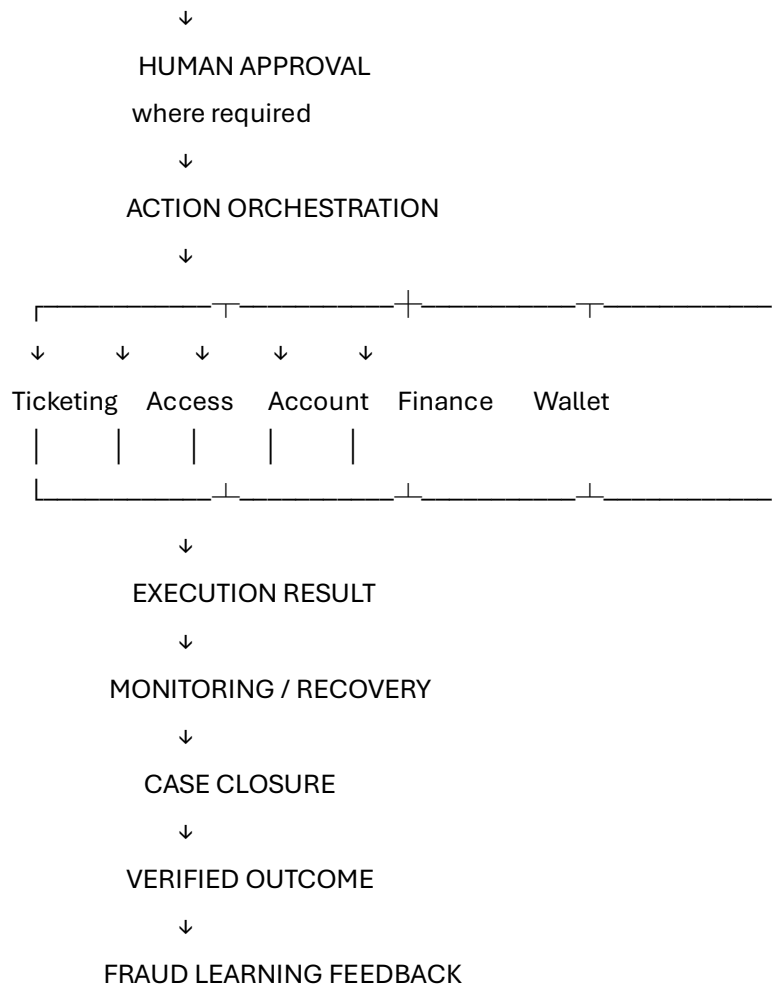
Actions

View Performance | Generate Report | Review Patterns | Send Optimization Feedback

Critical End-to-End Fraud Architecture

With Board 3, the complete Fraud module should operate like this:





One Case — Multiple Fraud Types

This is important.

Do **not** create separate investigations simply because the same fraud appears in multiple modules.

Example:

A suspicious payment occurs.

↓

Tickets are purchased.

↓

Tickets are transferred.

↓

Credentials appear on several devices.

↓

Multiple invalid gate scans occur.

↓

Refund requested.

This should potentially become:

One Fraud Case

containing:

- Payment Fraud
- Ticket Transfer Abuse
- Credential Sharing
- Access Abuse
- Refund Abuse

rather than five disconnected cases.

Parent / Child Cases

For very large fraud networks, support:

Master Case

→ Child Investigation 1

→ Child Investigation 2

→ Child Investigation 3

Example:

Master Case

Unauthorized Reseller Network

Child cases:

- Payment cluster
- Ticket-transfer cluster
- Membership abuse
- Access abuse

This allows large investigations without making one case impossible to manage.

Alert ≠ Case ≠ Fraud

Soft Labs should preserve this distinction throughout the UI and database.

Alert

Something suspicious was detected.

Case

The suspicious activity requires investigation.

Confirmed Fraud

The investigation concluded fraud occurred.

These should never be treated as the same object.

Case Object

I recommend a standardized object:

FraudCase

|

└─ Case ID

└─ Tenant

└─ Venue

└─ Status

└─ Priority

└─ Risk

└─ Fraud Type(s)

|

└─ Primary Entity

└─ Related Entities

└─ Related Alerts

└─ Related Assessments

|

└─ Assigned Investigator

└─ Team

└─ SLA

|

└─ Estimated Exposure

└─ Confirmed Loss

└─ Prevented Value

└─ Recovered Value

|

└─ Evidence

└─ Timeline

└─ Relationship Graph

└─ Investigation Notes

|

└─ Decision

└─ Disposition

└─ Recommended Actions

└─ Approvals

└─ Execution Results

|

└─ Monitoring Period

└─ Resolution

└— Closure Reason

|
└— Audit History

Evidence Must Reference Source — Not Copy Everything

Where possible, evidence should reference the authoritative source object.

Example:

Transaction Evidence

→ Payment Transaction ID

Ticket Evidence

→ Ticket ID

Scan Evidence

→ Access Scan ID

Refund Evidence

→ Refund ID

However, for material investigation evidence, TICVAI should preserve an immutable snapshot/reference sufficient to reconstruct what the investigator saw at the time.

This protects audit integrity if the operational object later changes.

Fraud Case Versioning

Material case decisions should be versioned.

Example:

Case Decision V1

Likely Fraud

↓

New evidence discovered

↓

Decision V2

Confirmed Fraud

The system should preserve both.

Do not overwrite the original conclusion.

Human Decision Is Mandatory for Serious Conclusions

AI can provide:

Recommended conclusion: High likelihood of coordinated fraud.

But for serious customer-impacting cases, I recommend the formal disposition remain:

Human Decision

especially when actions may involve:

- Account suspension
- Credential invalidation
- Refund withholding
- Membership restriction
- Reseller restriction

This aligns directly with the AI Governance module we already designed.

Explainability

The investigator should see structured evidence such as:

Risk increased because 24 accounts share five devices, 62 rapid ticket transfers occurred, and 31 related invalid access attempts were detected.

Not:

“The AI believes this person is fraudulent.”

And never expose hidden model chain-of-thought.

Case Risk vs Entity Risk

Another important distinction:

A case may be:

Critical

while individual entities have different risk levels.

Example:

Entity	Risk
Account A	Critical
Account B	High
Account C	Medium
Device A	Critical
Ticket A	High
Ticket B	Low

The investigator should decide which entities are actually relevant.

Do not automatically punish every entity connected to a high-risk cluster.

Recovery vs Prevention

TICVAI should distinguish:

Prevented Fraud

Fraudulent value stopped before loss.

Recovered Fraud

Loss occurred but money/value was later recovered.

Confirmed Loss

Fraud occurred and value was not recovered.

Example:

AED 100K Exposure

↓

AED 40K prevented

AED 20K recovered

AED 30K confirmed loss

AED 10K still under investigation.

This will give management much more meaningful fraud KPIs.

Feedback to Boards 1 & 2

Board 3 provides the **verified outcome**.

That is extremely valuable.

Board 1 predicted:

Transaction risk = 92.

Board 3 concludes:

Confirmed Fraud.

This becomes positive labeled evidence.

Another transaction:

Risk = 88.

Board 3:

Legitimate / False Positive.

That is equally valuable.

Therefore:

Detection → Investigation → Verified Outcome → Evaluation → Improvement

creates the learning loop.

Important Model Learning Boundary

Do not design:

Analyst clicks "Fraud" → AI automatically retrains itself tonight → new model automatically goes live.

Instead:

Verified Outcomes

↓

Training / Evaluation Dataset

↓

Candidate Model / Rule Improvement

↓

Backtesting

↓

Performance Comparison

↓

Governance Review

↓

Approved Deployment

This is much safer and more enterprise-ready.

Fraud Module — Final Architecture

With Board 3 complete, our full Fraud module is:

Board	Responsibility	Screens
1	Payment, Transaction, Refund & Chargeback Fraud	10
2	Ticket, Account, Credential & Behavioral Fraud	10
3	Fraud Investigation, Case Management & Response	10
TOTAL	AI Fraud & Risk Intelligence	30

The progression is very clean:

Board 1 — MONEY

Detect suspicious commercial activity.

↓

Board 2 — BEHAVIOR

Detect suspicious usage and relationships.

↓

Board 3 — INVESTIGATE & RESPOND

Connect evidence → investigate → decide → approve → execute through owning modules → recover → close → learn.

This completes the functional scope for **AI Fraud & Risk Intelligence**. After we **Design Board 3**, the next major AI module is the final layer: **Core AI Platform & Enterprise Intelligence**.